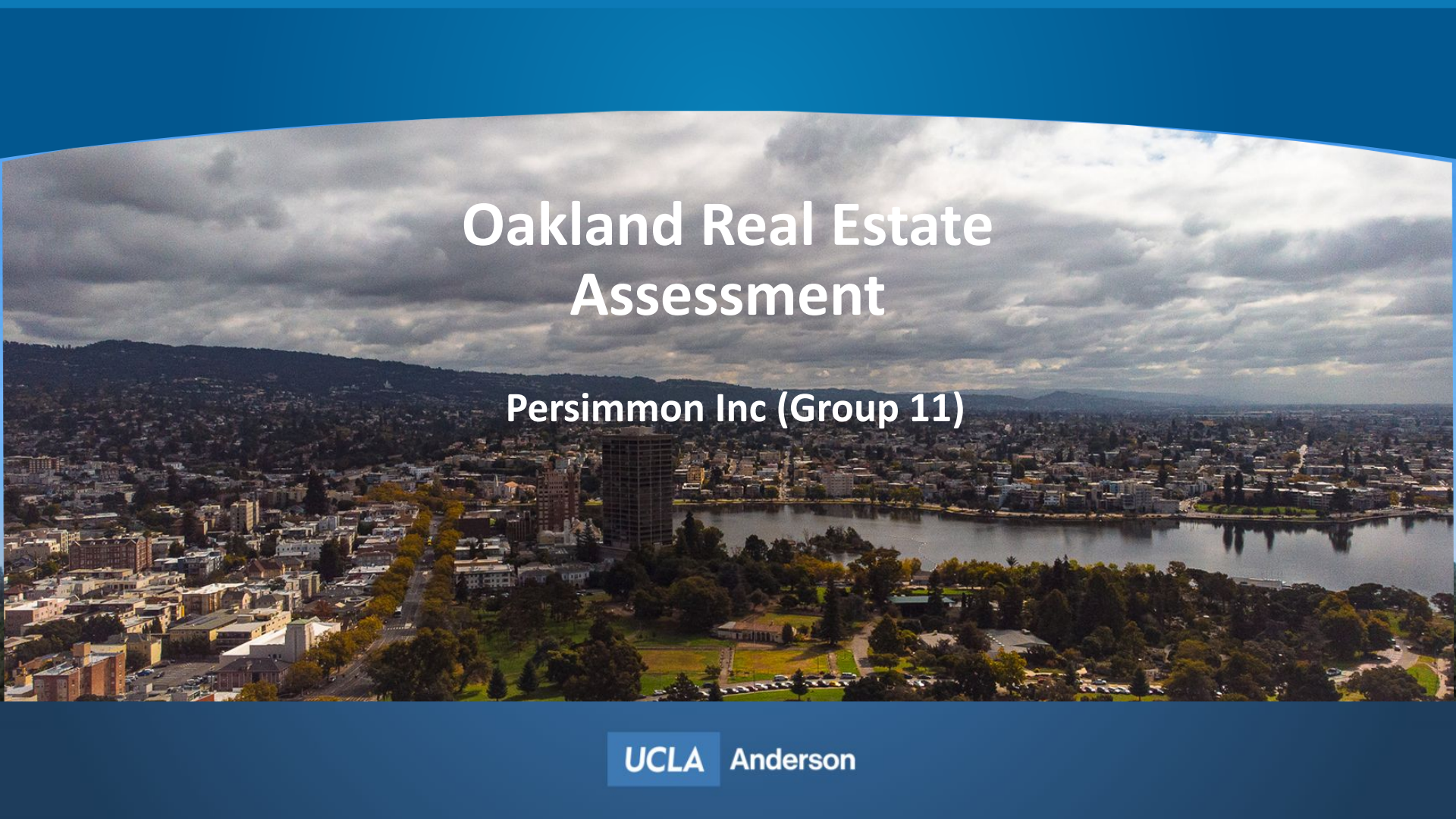
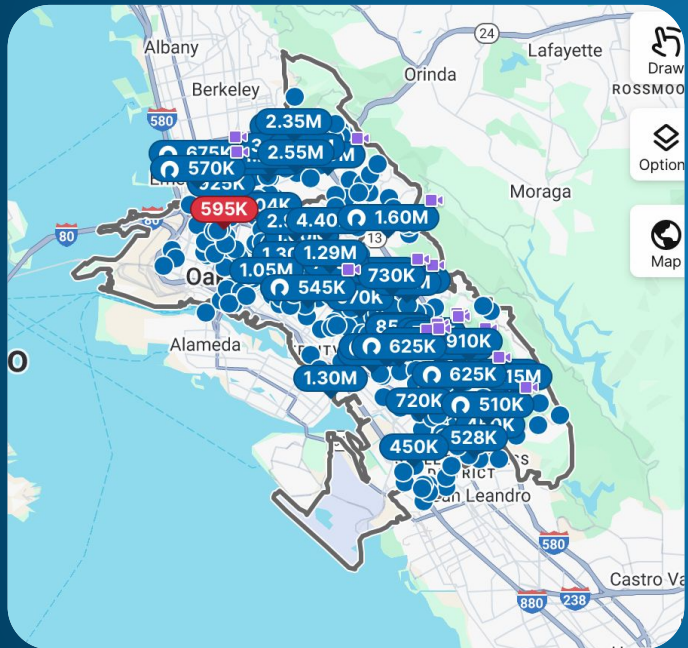




Oakland Real Estate Assessment

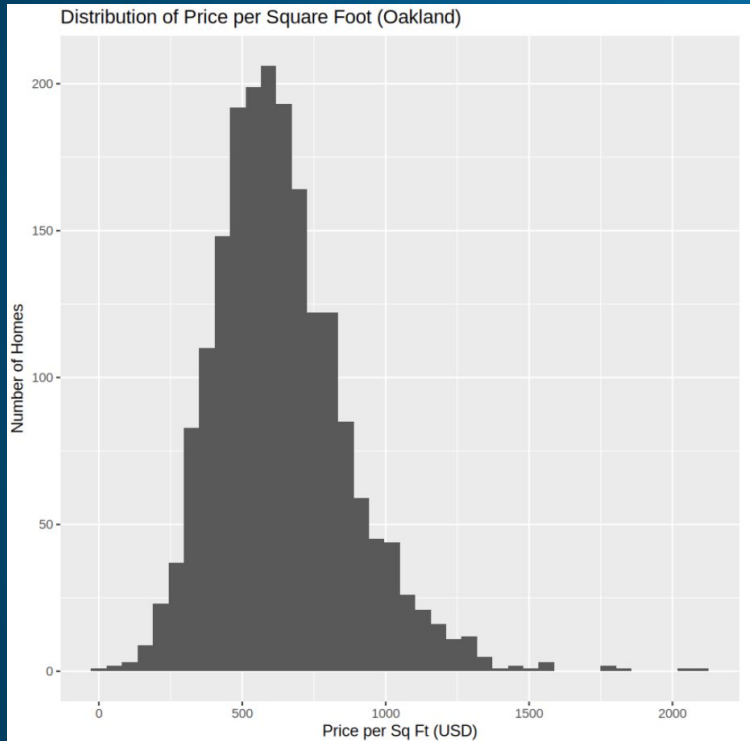
Persimmon Inc (Group 11)

Oakland Home Sales Data Set



- After data cleaning, we focused on 1,956 listings located within the official Oakland city limits.
- Models 1–2 are estimated on homes with non-missing price, square footage, beds, and baths ($N \approx 1,938$ homes)
- Model 3 uses a smaller subset ($N = 1,621$) with additional required data on HOLC grade, amenities, ZIP, and sale year

Price per Square Foot in Oakland



The average price per square foot among sold homes over the past 5 years is \$637, with most homes clustered between \$400–\$800 per square foot.

Model 1 - Price vs. Square Feet

The intercept of \$242,100 represents the predicted price of a home with zero square feet

```
Call:
lm(formula = PRICE ~ SQUARE.FEET, data = oakland_homes)

Residuals:
      Min       1Q   Median       3Q      Max
-3801447 -246827  -76955   187079  3800916

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.421e+05  2.053e+04   11.79  <2e-16 ***
SQUARE.FEET  4.820e+02  9.718e+00   49.60  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 493900 on 1948 degrees of freedom
Multiple R-squared:  0.5581,    Adjusted R-squared:  0.5578
F-statistic: 2460 on 1 and 1948 DF,  p-value: < 2.2e-16
```

Model 1 - Price vs. Square Feet

For each additional square foot, the model predicts an average price increase of approximately \$482

```
Call:
lm(formula = PRICE ~ SQUARE.FEET, data = oakland_homes)

Residuals:
    Min       1Q   Median       3Q      Max
-3801447 -246827  -76955   187079  3800916

Coefficients:
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

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F-statistic: 2460 on 1 and 1948 DF,  p-value: < 2.2e-16
```

Model 1 - Price vs. Square Feet

The extremely small p-value ($< 2.2e-16$) confirms this effect is highly statistically significant.

```
Call:
lm(formula = PRICE ~ SQUARE.FEET, data = oakland_homes)

Residuals:
    Min       1Q   Median       3Q      Max
-3801447 -246827  -76955   187079  3800916

Coefficients:
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(Intercept)  2.421e+05  2.053e+04  11.79  <2e-16 ***
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

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F-statistic: 2460 on 1 and 1948 DF,  p-value: < 2.2e-16
```



Model 2 - Price vs. Square Feet + Bed + Baths

Comparison of Simple and Extended Price Models

Dependent variable: Sale Price (price)

	Size Only (1)	Size + Beds + Baths (2)
SQUARE.FEET	481.978*** (9.718)	570.244*** (20.409)
BEDS		-68,163.100*** (12,322.810)
BATHS		57,892.000*** (17,668.910)
Constant	242,149.700*** (20,534.490)	173,059.400*** (25,308.800)
Observations	1,950	1,938
R2	0.558	0.578
Adjusted R2	0.558	0.577

Note: *p<0.1; **p<0.05; ***p<0.01

After controlling for beds and baths, the value of each extra square foot increases.

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Holding size and baths constant, **more bedrooms are associated with a lower sale price**, suggesting buyers prefer fewer, larger rooms.

Model 2 - Price vs. Square Feet + Bed + Baths

Comparison of Simple and Extended Price Models

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	Size Only (1)	Size + Beds + Baths (2)
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Holding size and bedrooms constant, **more bathrooms are associated with a higher sale price**, suggesting buyers are willing to pay a premium for extra bathrooms.

Model 2 - Price vs. Square Feet + Bed + Baths

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	Size Only (1)	Size + Beds + Baths (2)
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Overall, the extended model fits the data better (higher R²)

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Note: *p<0.1; **p<0.05; ***p<0.01

Overall, the extended model fits the data better (higher R²)

Model 2 - Predicting Typical Home Price

We consider a 2-bedroom, 2-bath home with 1,500 square feet:

$\text{PRICE} = 1.7305940 + 0.0057024 * 1500 - 0.6816310 * 2 + 0.5789200 * 2 = 10.078772$,
meaning the predicted price is **\$1,007,884**.

The 95% confidence interval is [\$979,812 \$1,035,956]

```
Call:
lm(formula = PRICE ~ SQUARE.FEET + BEDS + BATHS, data = homes_df)

Residuals:
    Min       1Q   Median       3Q      Max
-46.047  -2.414   -0.590    1.801   33.932

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.7305940   0.2530880    6.838 1.07e-11 ***
SQUARE.FEET  0.0057024   0.0002041   27.940 < 2e-16 ***
BEDS        -0.6816310   0.1232281   -5.531 3.61e-08 ***
BATHS        0.5789200   0.1766891    3.276 0.00107 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.716 on 1934 degrees of freedom
Multiple R-squared:  0.5779,    Adjusted R-squared:  0.5772
F-statistic: 882.5 on 3 and 1934 DF,  p-value: < 2.2e-16
Regression equation:
PRICE = 1.731 + 0.006 * SQUARE.FEET + -0.682 * BEDS + 0.579 * BATHS
```

$\text{PRICE} = 1.7305940 + 0.0057024 * \text{square feet} - 0.6816310 * \text{bedrooms} + 0.5789200 * \text{bathrooms}$

Model 2 - Predicting Typical Home Price

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-46.047  -2.414   -0.590    1.801   33.932

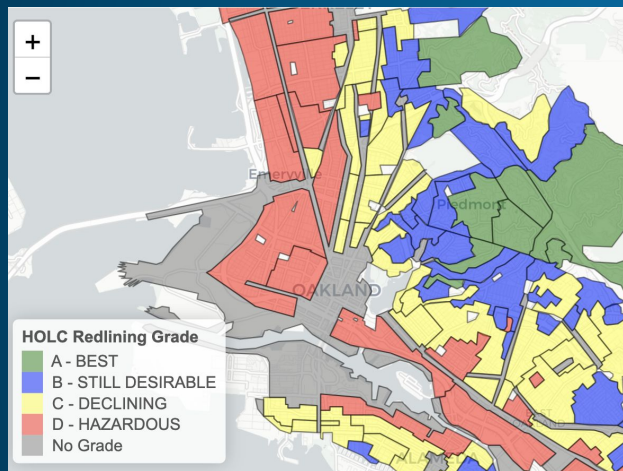
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.7305940   0.2530880    6.838 1.07e-11 ***
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```

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Model 3 - Extended Model

+HOLC grade, ZIP, amenities, sale year



Historical redlining and neighborhoods



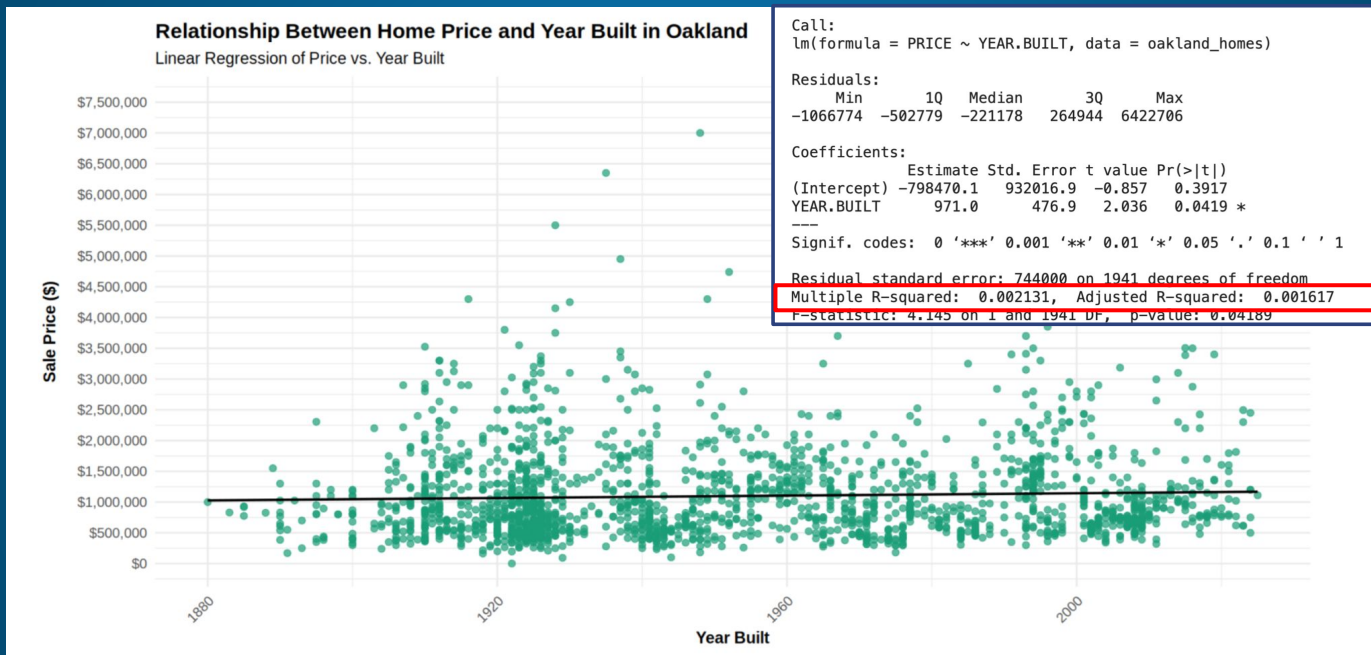
Distance to quality schools, job geography, amenities



Sale Year

Model 3 - Extended Model

Age is Just a Number*



*Statistically significant, but explains almost 0% of price variation

Model 3 - Extended Model

Key Observations

Historic HOLC grade

- HOLC B homes sell for about **\$80k more** than similar **HOLC C** homes (after controls).

School quality access

- Within **0.5 miles** of a quality school → about **\$120k higher** sale price.

Job geography

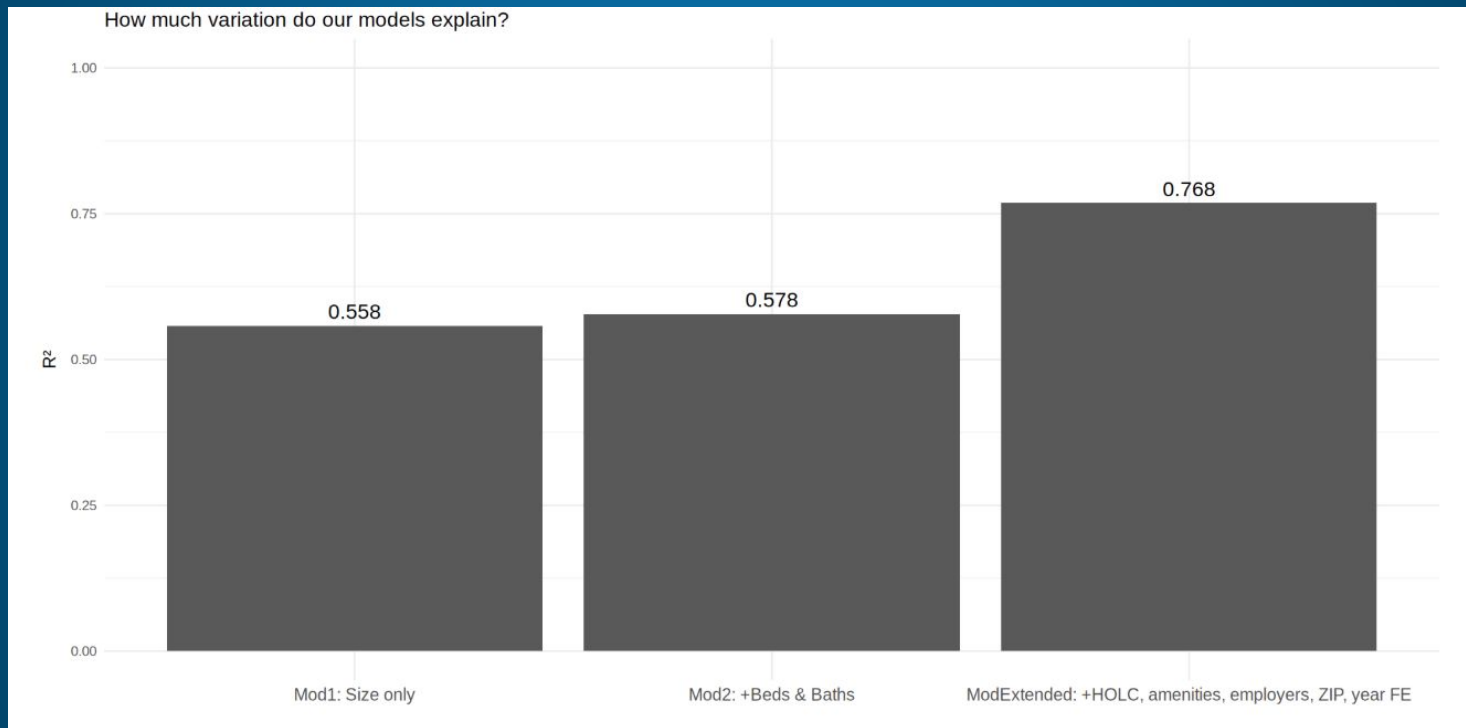
- **1 mile closer to Pixar/Emeryville** → roughly **+\$300k**
- Priciest homes are **farther from downtown and industrial areas** → **+\$264-584k per mile** further.

Amenities & Waterfront

- Close to **grocery + retail/dining** → about **+\$180–200k**.
- **Waterfront** has the biggest premium: roughly **+\$380k per mile** closer.

Model 3 - Extended Model

Key Observations



Model 3 - Extended Model

Identifying Investment Opportunities

Recommendations

- Target historically HOLC B neighborhoods (and select C/D blocks) that have strong present-day fundamentals (clean employers, good amenities)
- Prioritize neighborhoods within easy reach of quality schools if willing to pay for premium.
- **Avoid homes right next to heavy industrial/port areas** – buyers pay a clear premium to be farther away, so these locations sell at a discount.
- **Scan ZIP codes with premiums/discounts of about $\pm\$200k$ vs. 94609** – these “in-between” areas often look like emerging neighborhoods in transition or early gentrification.



A Closer Look at Key Areas

- 94607 (West Oakland / Jack London): **prices are offset by strong location advantages.**
- 94608 (North Oakland / Emeryville): **market is driven by major attractors like Pixar.**

How to Use our Models

The model can help identify high-ROI homes using 2 strategies:

(1) Undervaluation / Mispricing with Model 3

- Perform a regression capturing recently-closed sales to estimate fair market value based on the number of beds, baths, and square feet
- Assess current listings to identify homes with a >10% delta between list price and the modeled fair value.
- Analyze each highlighted listing to ensure it isn't a false signal or outlier.

(2) Value-Add / Forced Equity with Model 2

- Perform a regression to identify fair value based on bathrooms, bedrooms, or square footage
- Quantify the modeled price opportunity from adding a bathroom, bedroom, or expanding square footage
- Identify opportunities where fair value increase is significantly higher than cost to add beds/baths/sqft



Let's Invest Together!



Appendix

Effect of Lot Size on Non-SFR Homes

Single-family homes are about \$275K more expensive than the baseline (controlling for size and lot).

Townhouses are about \$121K more expensive than the baseline.

```
Call:
lm(formula = PRICE ~ SQUARE.FEET + LOT.SIZE + PROPERTY.TYPE,
    data = oakland_homes)

Residuals:
    Min       1Q   Median       3Q      Max
-3944510 -267238  -60572   216120  3641437

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  5.473e+03  5.032e+04   0.109  0.9134
SQUARE.FEET  4.907e+02  1.060e+01  46.278 < 2e-16
LOT.SIZE     1.715e+00  8.139e-01   2.107  0.0353
PROPERTY.TYPEMulti-Family (2-4 Unit) -1.263e+05  1.337e+05  -0.944  0.3451
PROPERTY.TYPEMulti-Family (5+ Unit) -2.191e+06  3.049e+05  -7.187  9.8e-13
PROPERTY.TYPEOther -8.950e+05  5.019e+05  -1.783  0.0747
PROPERTY.TYPESingle Family Residential 2.749e+05  4.852e+04   5.667  1.7e-08
PROPERTY.TYPETownhouse 1.211e+05  5.675e+04   2.133  0.0330

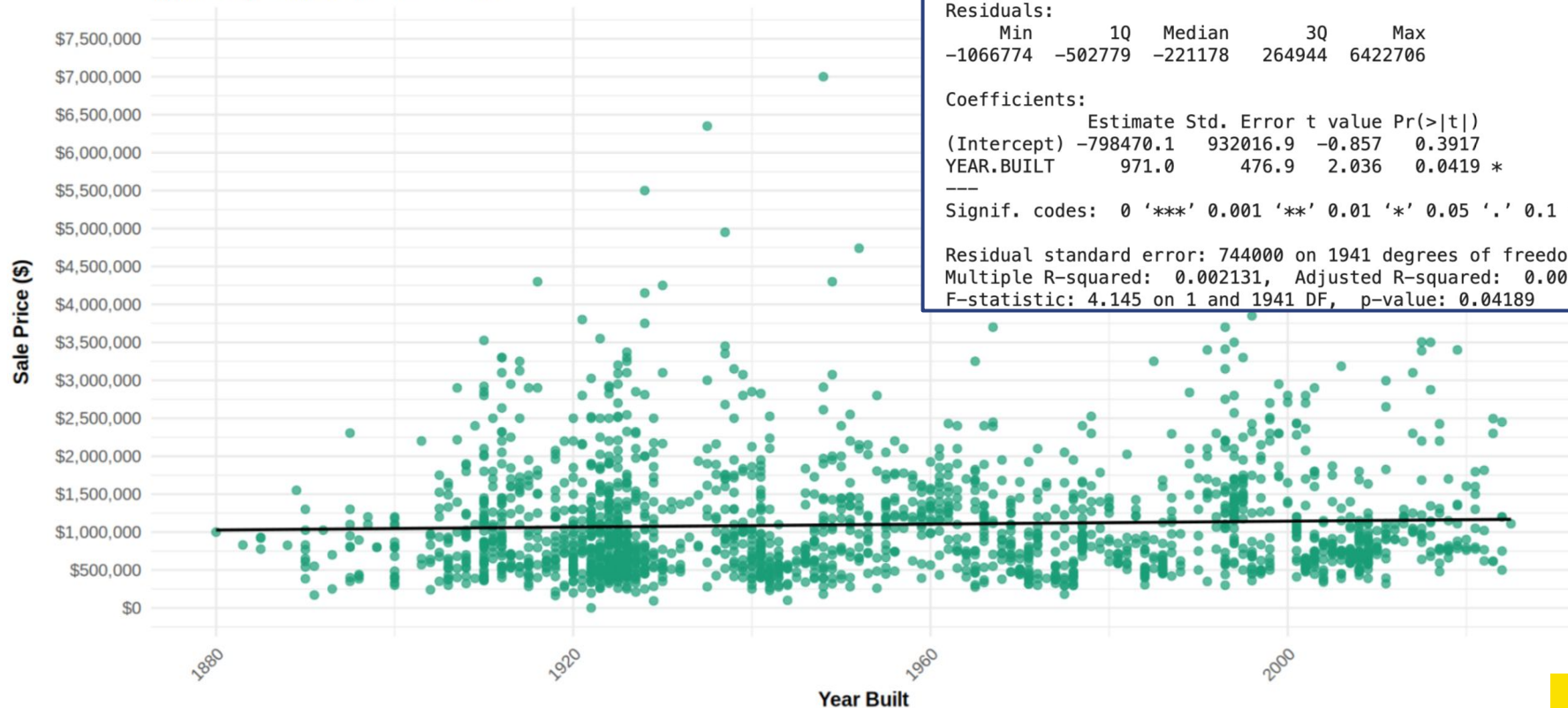
(Intercept)
SQUARE.FEET ***
LOT.SIZE *
PROPERTY.TYPEMulti-Family (2-4 Unit)
PROPERTY.TYPEMulti-Family (5+ Unit) ***
PROPERTY.TYPEOther .
PROPERTY.TYPESingle Family Residential ***
PROPERTY.TYPETownhouse *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5e+05 on 1730 degrees of freedom
(212 observations deleted due to missingness)
Multiple R-squared:  0.5764,    Adjusted R-squared:  0.5747
F-statistic: 336.3 on 7 and 1730 DF,  p-value: < 2.2e-16
```

Remember: Age is Just a Number

Relationship Between Home Price and Year Built in Oakland

Linear Regression of Price vs. Year Built



Model 3 - Extended Model

Key Observations

```
Call:
lm(formula = price_num ~ SQUARE.FEET + BEDS + BATHS + is_HOLC_A +
    is_HOLC_B + is_HOLC_D + within_0_5_mile_quality_school +
    distance_to_bart_miles + dist_to_employer_pixar_miles + dist_to_employer_downtown_miles +
    dist_to_employer_industrial_miles + dist_to_employer_college_miles +
    dist_to_grocery_miles + dist_to_retail_dining_miles + dist_to_shopping_center_miles +
    dist_to_hiking_miles + dist_to_waterfront_miles + zip_factor +
    sale_year_fe, data = oakland_ext)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-2989523	-182256	-9625	163769	3077364

Coefficients:

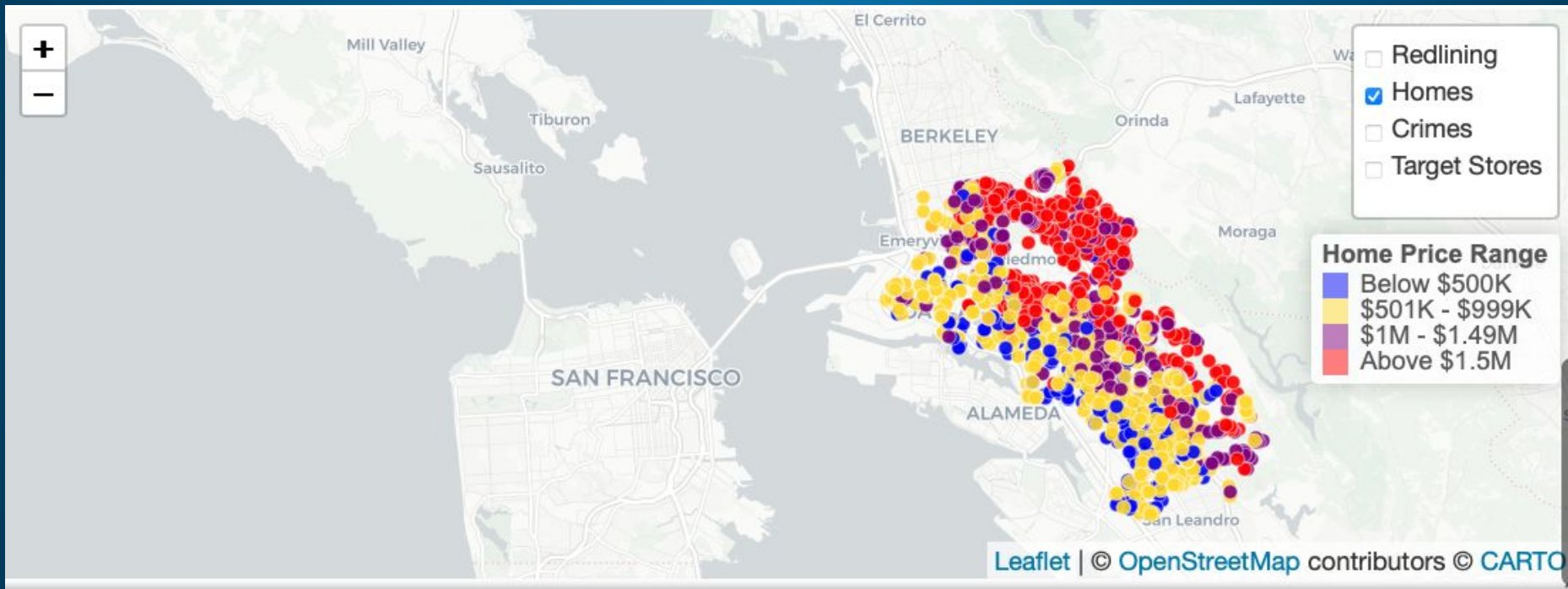
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-532629.8	247789.9	-2.150	0.031744 *
SQUARE.FEET	526.9	19.4	27.156	< 2e-16 ***
BEDS	-26424.5	11744.2	-2.250	0.024585 *
BATHS	15433.0	16104.7	0.958	0.338062
is_HOLC_A	33710.3	40146.7	0.840	0.401215
is_HOLC_B	82099.3	26351.0	3.116	0.001869 **
is_HOLC_D	-29918.8	35196.1	-0.850	0.395419
within_0_5_mile_quality_school	124694.2	23305.7	5.350	1.01e-07 ***
distance_to_bart_miles	10423.2	34825.1	0.299	0.764750
dist_to_employer_pixar_miles	-311526.0	72663.2	-4.287	1.92e-05 ***
dist_to_employer_downtown_miles	587240.4	106906.7	5.493	4.60e-08 ***
dist_to_employer_industrial_miles	264291.6	77384.2	3.415	0.000653 ***
dist_to_employer_college_miles	199700.6	57627.3	3.465	0.000544 ***
dist_to_grocery_miles	-199594.9	52139.8	-3.828	0.000134 ***
dist_to_retail_dining_miles	-183961.5	46862.9	-3.926	9.03e-05 ***
dist_to_shopping_center_miles	-7877.0	40605.9	-0.194	0.846212
dist_to_hiking_miles	57301.4	43794.8	1.308	0.190926
dist_to_waterfront_miles	-381675.1	108424.8	-3.520	0.000443 ***

zip_factor94501	-211650.1	375968.4	-0.563	0.573551
zip_factor94560	402506.5	377798.1	1.065	0.286857
zip_factor94601	58556.9	119494.3	0.490	0.624174
zip_factor94602	286778.3	98085.2	2.924	0.003507 **
zip_factor94603	151414.2	156983.7	0.965	0.334932
zip_factor94605	249015.9	135821.1	1.833	0.066929 .
zip_factor94606	-47549.1	103694.5	-0.459	0.646621
zip_factor94607	81473.3	119115.7	0.684	0.494085
zip_factor94608	-217837.7	118433.5	-1.839	0.066055 .
zip_factor94610	422247.8	93457.5	4.518	6.70e-06 ***
zip_factor94611	272840.6	75135.8	3.631	0.000291 ***
zip_factor94612	302547.2	114663.2	2.639	0.008407 **
zip_factor94618	537437.1	71285.3	7.539	7.92e-14 ***
zip_factor94619	273123.0	122465.6	2.230	0.025874 *
zip_factor94621	249501.6	155373.9	1.606	0.108514
sale_year_fe2021	150076.5	43421.1	3.456	0.000562 ***
sale_year_fe2022	203580.0	44945.6	4.529	6.36e-06 ***
sale_year_fe2023	114232.4	46976.6	2.432	0.015139 *
sale_year_fe2024	96792.0	47237.2	2.049	0.040621 *
sale_year_fe2025	-15757.2	47359.0	-0.333	0.739391

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 351500 on 1583 degrees of freedom
 (317 observations deleted due to missingness)
 Multiple R-squared: 0.7676, Adjusted R-squared: 0.7622
 F-statistic: 141.3 on 37 and 1583 DF, p-value: < 2.2e-16

Homes sold in the last 5 years



Question 6

$$\text{PRICE} = 1.7305940 + 0.0057024 * \text{square feet} - 0.6816310 * \text{bedrooms} + 0.5789200 * \text{bathrooms}$$

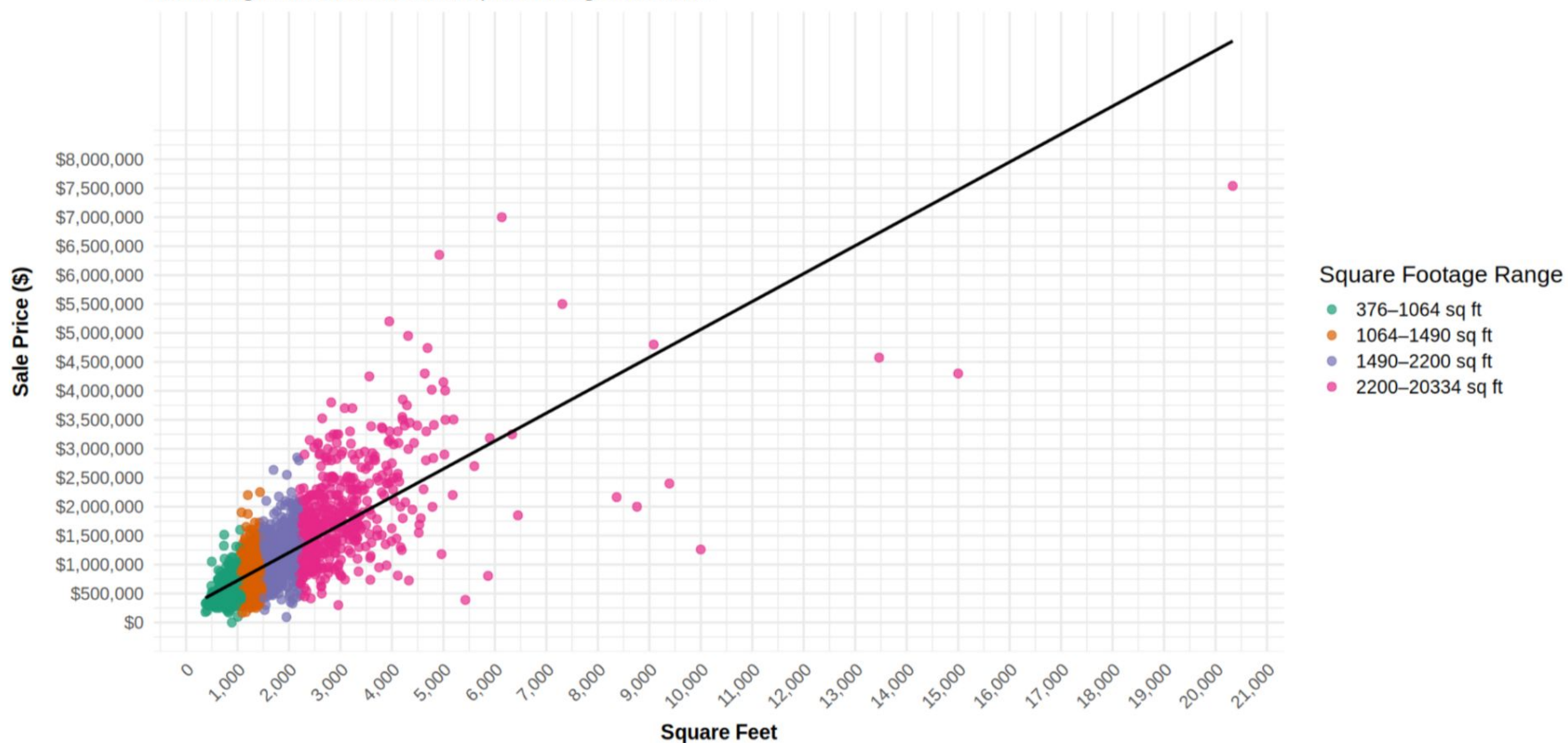
The 95% confidence interval is (9.79812, 10.35956)

A matrix: 1 × 3 of type dbl

	fit	lwr	upr
1	10.07884	9.79812	10.35956

Oakland Home Prices vs. Square Feet

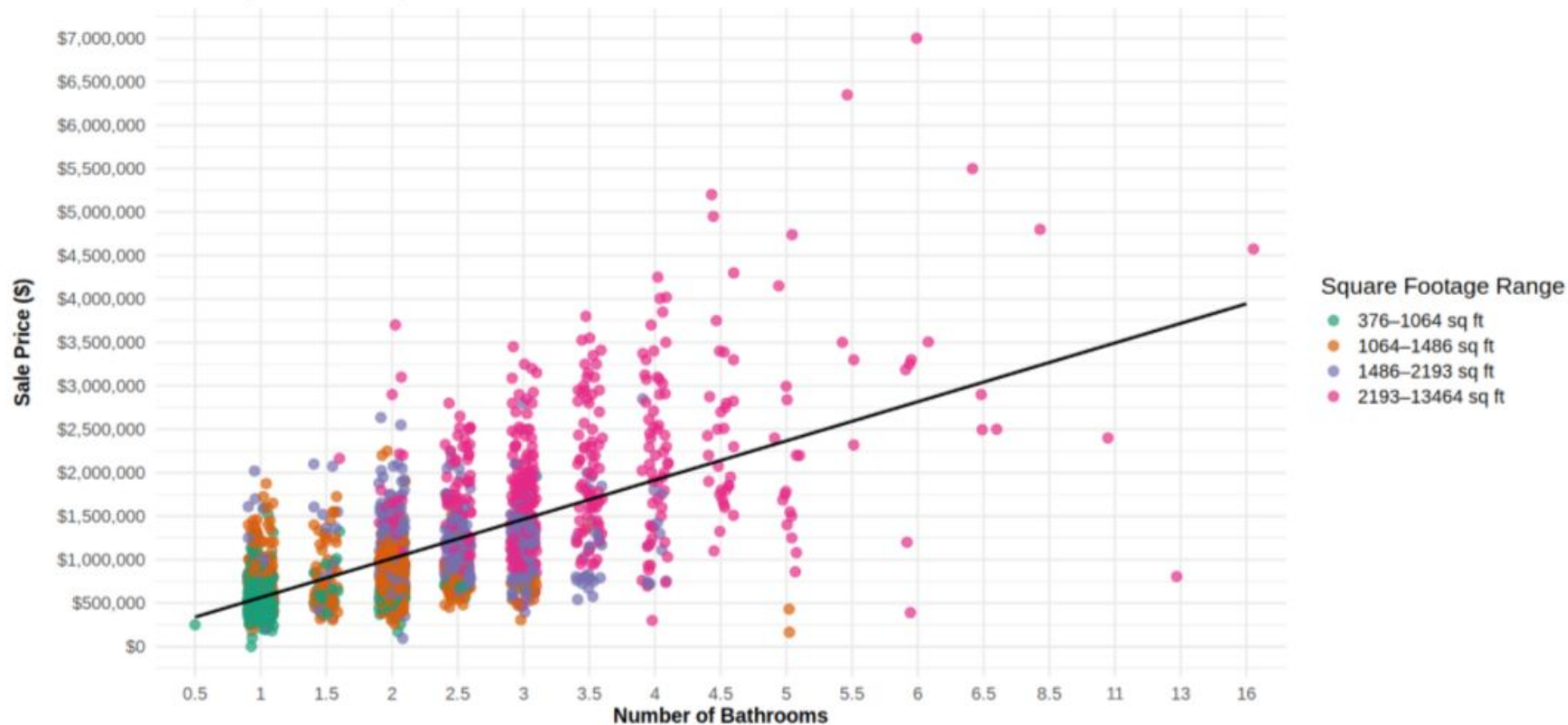
Linear Regression with Price and Square Footage Increments



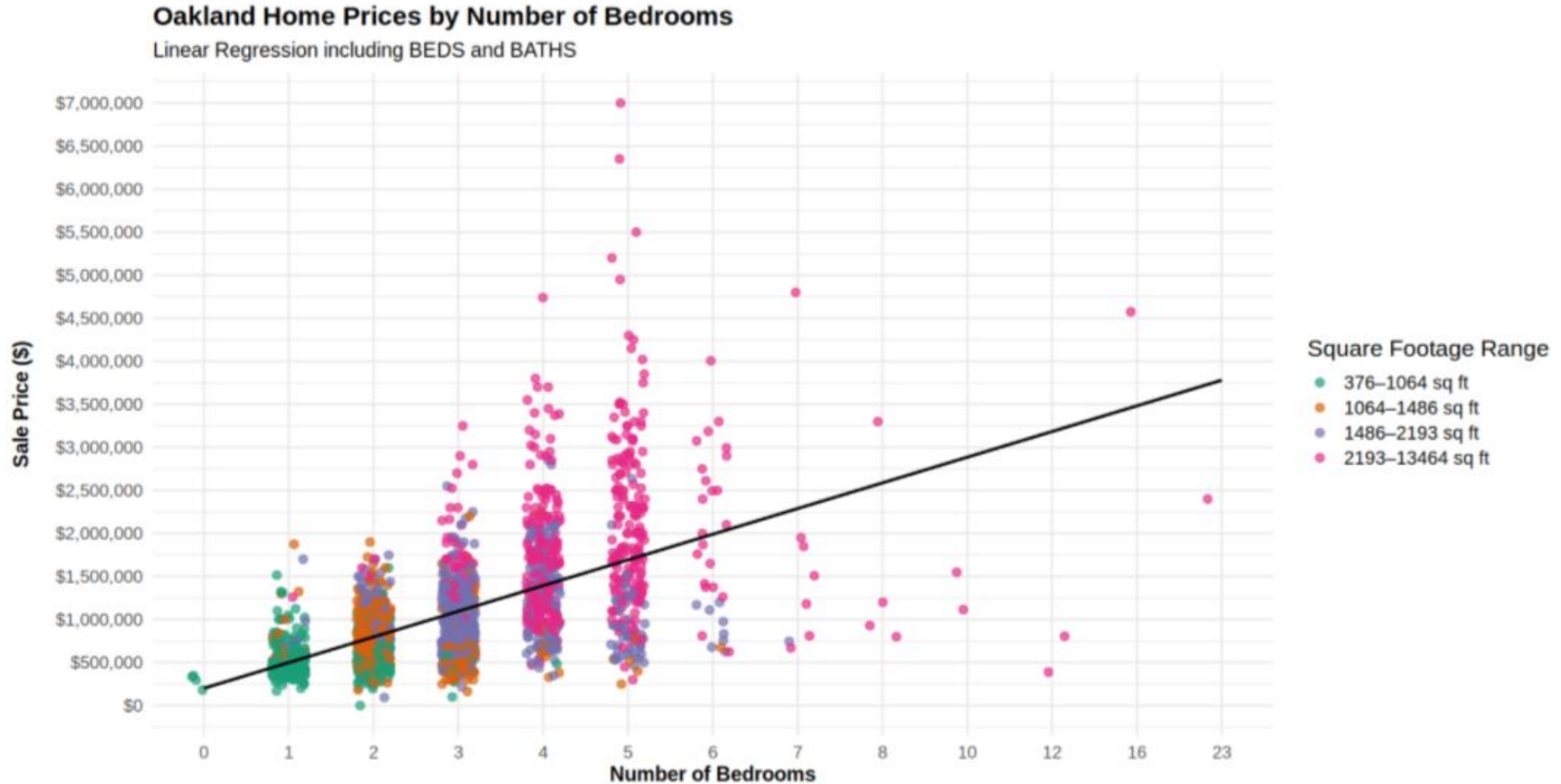
$$\text{PRICE} = 173,059.40 + 570.24 \times \text{SQUARE.FEET} - 68,163.10 \times \text{BEDS} + 57,892 \times \text{BATHS}.$$

Oakland Home Prices by Number of Bathrooms

Linear Regression including BEDS and BATHS



$$\text{PRICE} = 173,059.40 + 570.24 \times \text{SQUARE.FEET} - 68,163.10 \times \text{BEDS} + 57,892 \times \text{BATHS}.$$

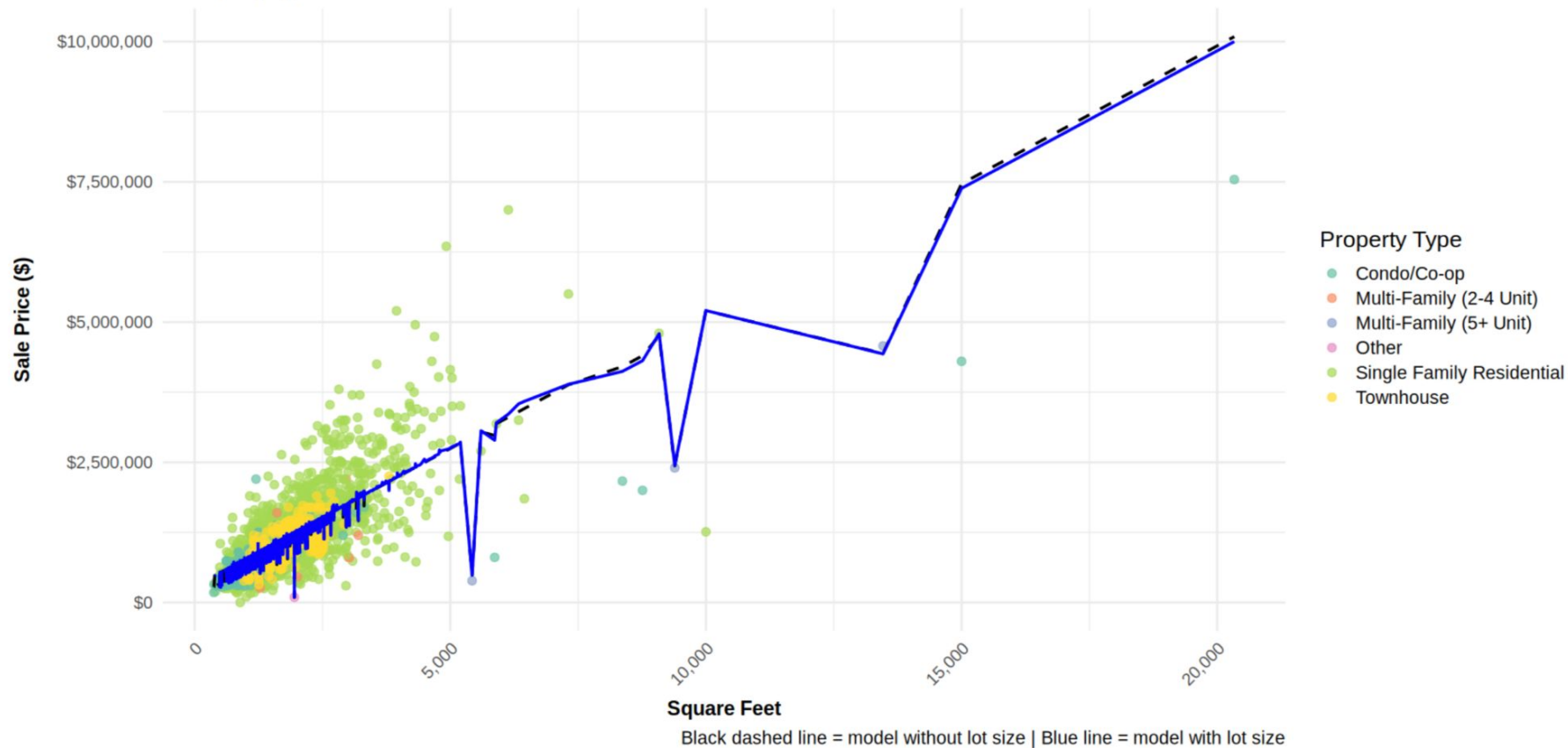


Question 4

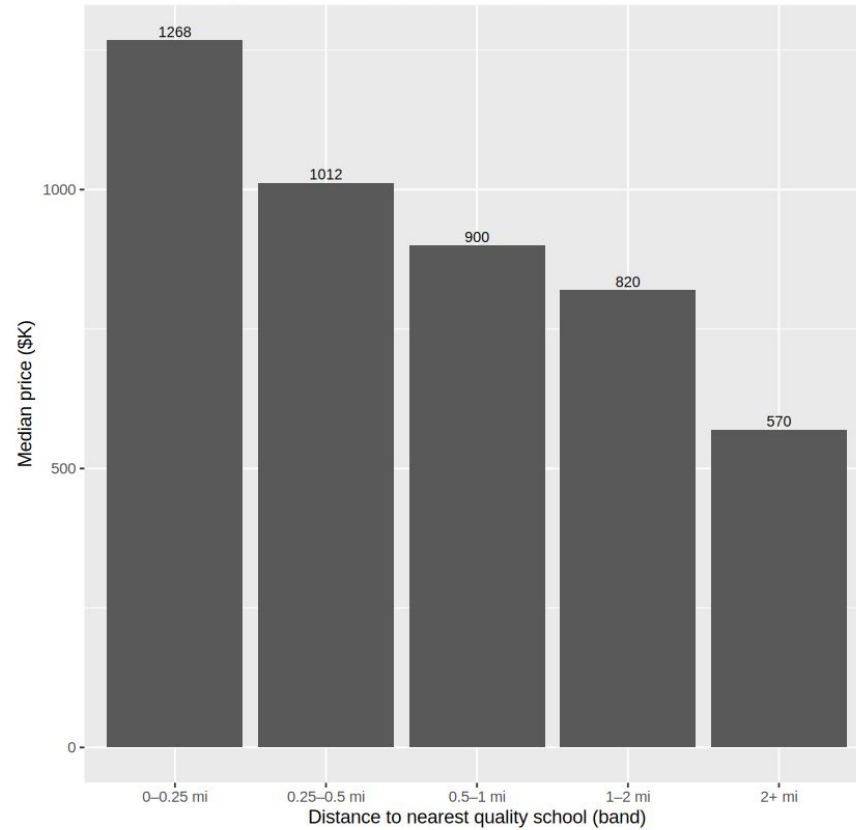


Lot Size and Property Type on Price Prediction

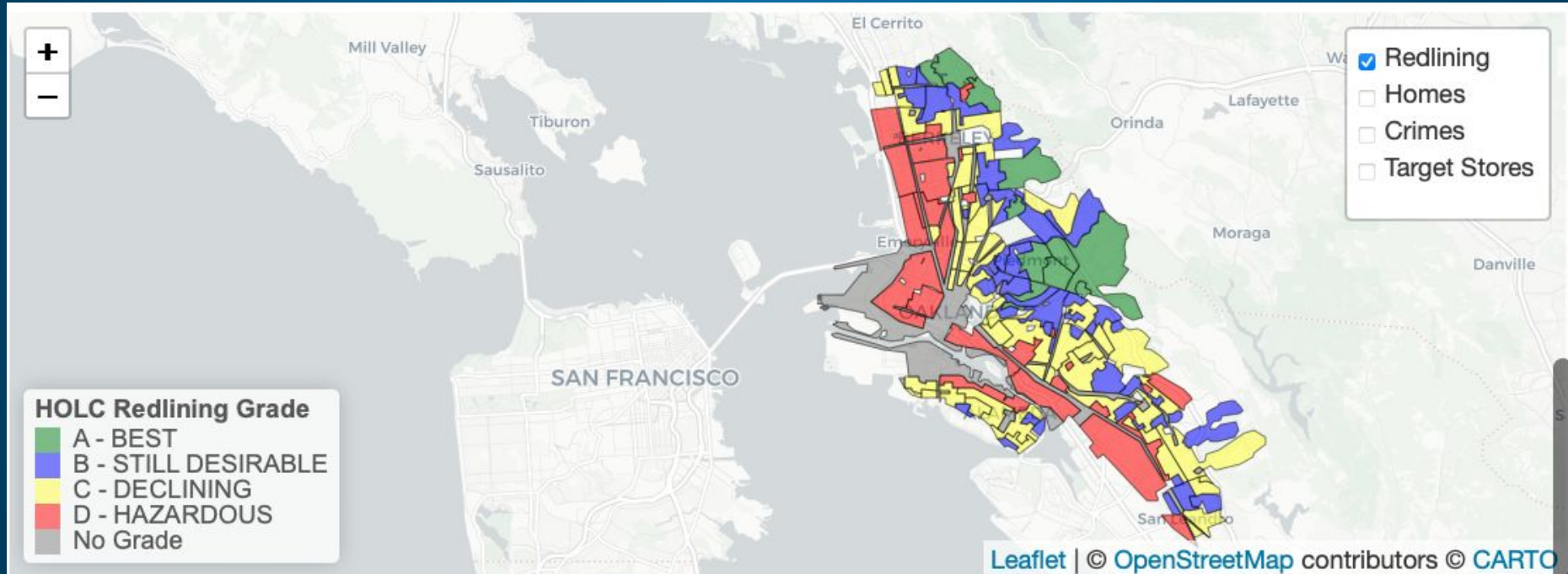
All Property Types in Oakland



Median home price by distance to nearest quality school



Redlining HOLC Grade



Crimes in the last 14 days

